

Transition Based Parsing: Learning

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Classifier-Based Parsing

Data-driven deterministic parsing:

- Deterministic parsing requires an **oracle**.
- An oracle can be approximated by a **classifier**.
- A classifier can be trained using **treebank** data.

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Three issues

- How to represent configurations by feature vectors?
- How to derive training data from treebanks?
- How to learn classifiers?

A feature representation $f(c)$ of a configuration c is a vector of simple features $f_i(c)$.

Typical Features

- Nodes:
 - ▶ Target nodes (top of S , head of B)
 - ▶ Linear context (neighbors in S and B)
 - ▶ Structural context (parents, children, siblings in G)

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- Attributes:
 - ▶ Word form (and lemma)
 - ▶ Part-of-speech (and morpho-syntactic features)
 - ▶ Dependency type (if labeled)
 - ▶ Distance (between target tokens)

Deterministic Parsing

To guide the parser, a linear classifier can be used:

$$t^* = \arg \max_t w \cdot f(c, t)$$

Weight vector w learned from treebank data.

Using classifier at run-time

```
PARSE( $w_1, \dots, w_n$ )  
1    $c \leftarrow ([ ]_S, [w_1, \dots, w_n]_B, \{ \})$   
2   while  $B_c \neq [ ]$   
3      $t^* \leftarrow \arg \max_t w \cdot f(c, t)$   
4      $c \leftarrow t^*(c)$   
5   return  $T = (\{w_1, \dots, w_n\}, A_c)$ 
```

- Training instances have the form $(f(c), t)$, where
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 - ▶ $f(c)$ is a feature representation of a configuration c ,
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- Given a dependency treebank, we can sample the oracle function o as follows:
 - ▶ For each sentence x with gold standard dependency graph G_x , construct a transition sequence $C_{0,m} = (c_0, c_1, \dots, c_m)$ such that
$$c_0 = c_s(x),$$
$$G_{c_m} = G_x$$
 - ▶ For each configuration $c_i (i < m)$, we construct a training instance $(f(c_i), t_i)$, where $t_i(c_i) = c_{i+1}$.

Standard Oracle for Arc-Eager Parsing

$o(c, T) =$

- **Left-Arc** if $\text{top}(S_c) \leftarrow \text{first}(B_c)$ in T
- **Right-Arc** if $\text{top}(S_c) \rightarrow \text{first}(B_c)$ in T
- **Reduce** if $\exists w < \text{top}(S_c) : w \leftrightarrow \text{first}(B_c)$ in T
- **Shift** otherwise

Online Learning with an Oracle

```
LEARN( $\{T_1, \dots, T_N\}$ )
1    $w \leftarrow 0.0$ 
2   for  $i$  in  $1..K$ 
3     for  $j$  in  $1..N$ 
4        $c \leftarrow ([\ ]_S, [w_1, \dots, w_{n_j}]_B, \{\})$ 
5       while  $B_c \neq [\ ]$ 
6          $t^* \leftarrow \arg \max_t w.f(c, t)$ 
7          $t_o \leftarrow o(c, T_i)$ 
8         if  $t^* \neq t_o$ 
9            $w \leftarrow w + f(c, t_o) - f(c, t^*)$ 
10         $c \leftarrow t_o(c)$ 
11  return  $w$ 
```

Oracle $o(c, T_i)$ returns the optimal transition of c and T_i

Example

Consider the sentence, 'John saw Mary'.

- Draw a dependency graph for this sentence.
- Assume that you are learning a classifier for the data-driven deterministic parsing and the above sentence is a gold-standard parse in your training data. You are also given that *John* and *Mary* are 'Nouns', while the POS tag of *saw* is 'Verb'. Assume that your features correspond to the following conditions:
 - ▶ The stack is empty
 - ▶ Top of stack is Noun and Top of buffer is Verb
 - ▶ Top of stack is Verb and Top of buffer is Noun

Initialize the weights of all your features to 5.0, except that in all of the above cases, you give a weight of 5.5 to *Left-Arc*. Define your feature vector and the initial weight vector.

- Use this gold standard parse during online learning and report the weights after completing one full iteration of Arc-Eager parsing over this sentence.